Quantum Information w/ trapped ions.

Outline:

1. Overview of experimental approach and techniques.
2. Laser manipulation
3. Hamiltonian and key interactions.

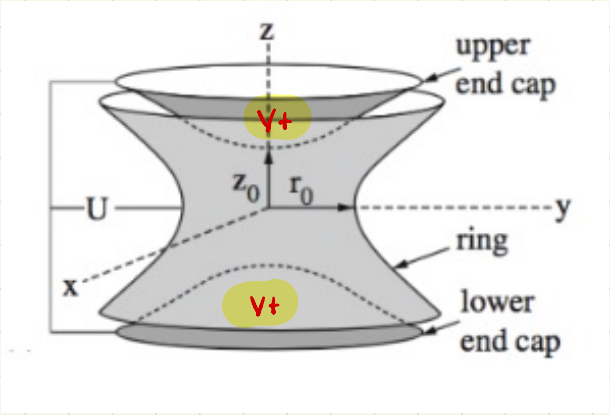
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① Trapping Ions:

- Similar setup to neutral atoms (source, vacuum system, etc.)
- Ionisation → via photoionisation.
- Trapping use interaction of ion w/ electric field.

Issue! Gauss's law: $\vec{\nabla} \cdot \vec{E} = 0$. (for source free region)

So a confining trap can only trap along two dimensions!



Form of the potential is

$$U(x,y,z) = \frac{1}{2} m\omega^2 \left(\frac{x^2+y^2}{2} - z^2 \right)$$

or similarly:

$$U(x,y,z) = \frac{1}{2} m\omega^2 \left(-\frac{x^2+y^2}{2} + z^2 \right)$$

But this will not trap! $\Longrightarrow$ anti trapping along at least 1 dimension.

Solution: Paul Trap (time averaged trap)

$$U(x,y,z) = \frac{1}{2} m\omega^2 \left(-\frac{x^2+y^2}{2} + z^2 \right) \cos(\omega_{RF} t)$$

where $\omega_{RF} >> \omega$ (fast modulation) (GHz)

Consider the motion of the ion (position $\vec{r}$)

$$\vec{r} = \vec{r}_f + \vec{r}_s$$

($\vec{r}_f$: $\sim \omega_{RF}$; $\vec{r}_s$: $<< \omega_{RF}$)

decompose in to fast & slow motion:

Ion slow motion is fixed on fast scale: (write as radial + z motion)

$$\ddot{\vec{r}}_f = -\frac{\omega^2}{2}\begin{pmatrix} -p_s \\ 2z_s \end{pmatrix} \sin(\omega_{RF} t) \quad \left(p_s = \sqrt{x_s^2 + y_s^2}\right) \Longrightarrow \vec{r}_f = \frac{\omega^2}{2\omega_{RF}^2}\begin{pmatrix} -p_s \\ 2z_s \end{pmatrix} \sin(\omega_{RF} t)$$

Slow motion obeys a time averaged equation:

$$\ddot{\vec{r}}_s = -\frac{\omega^2}{2}\left\langle \begin{pmatrix} -p_f \\ 2z_f \end{pmatrix} \sin(\omega_{RF} t) \right\rangle = -\frac{\omega^4}{4\omega_{RF}^2}\begin{pmatrix} p_s \\ 2z_s \end{pmatrix} \langle \sin^2(\omega_{RF} t) \rangle = -\frac{\omega^4}{8\omega_{RF}^2}\begin{pmatrix} p_s \\ 2z_s \end{pmatrix}$$

not $\langle \sin(\omega_{RF} t) \rangle = \frac{1}{2}$

Solution → harmonic motion at frequency $\frac{\omega^2}{2\sqrt{2}\,\omega_{RF}} = \nu$

(secular harmonic frequency)

where → $\omega \sim \frac{2z|e|U}{m}$

(z is ion charge)

Comments:

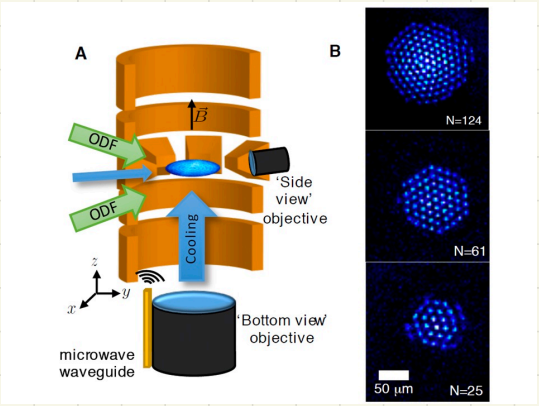
① Secular motion associated w/ ν occurs at frequencies reduced by $\sqrt{\omega_{RF}}$, but still relatively fast ($\sim 2\pi \times 10$ MHz)

② Fast motion at driving frequency occurs (micro motion).

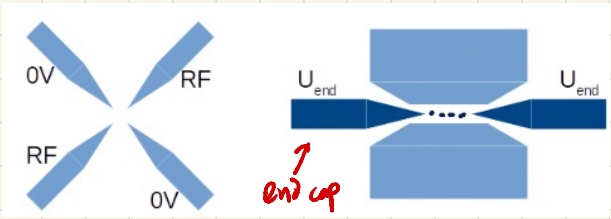
→ Can ignore this if ion is sufficiently cooled.

Trap configurations

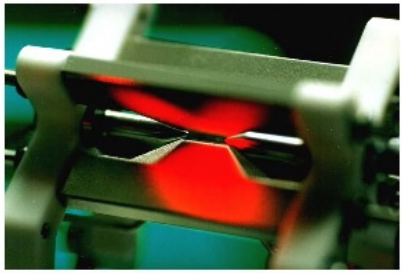
Penning trap (add magnetic field)



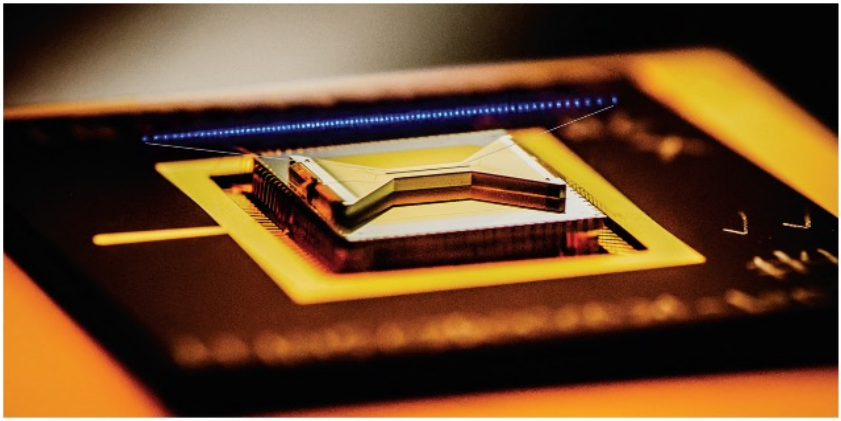
Linear Paul Trap (RF)



Innsbruck Linear Paul Trap



Surface trap



C. Monroe Science 364, 440 (2019)

Quantum mechanical Treatment (1D linear Paul trap)

* Harmonic trap results from time average (ignore micromotion)

Operator Hamiltonian:

two level atom

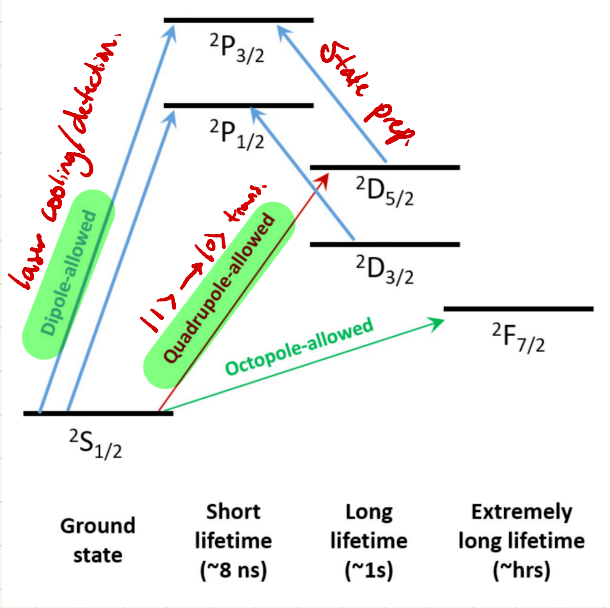
applied fields

$$H = \hbar\nu \hat{a}^\dagger \hat{a} + \hbar\omega_0 \sigma_z + H_{int}(\omega)$$

$\hbar\nu \hat{a}^\dagger \hat{a}$: harmonic motion at secular frequency

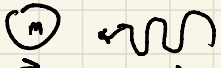
$H_{int}(\omega)$: interaction of system w/ external field (as per Rydberg atoms)

Aside on ion structure: (simplified)



Steps:

① Laser cooling. Ion initial temp $\sim 10{,}000\,K$ (2x surface of sun!!)



$\vec{p} = m\vec{v}$ $\vec{p} = \hbar\vec{k}$; $\omega_L = \omega_0 - \Delta$ $\omega_{L,atom} = \left(1 + \frac{v}{c}\right)\omega_L$

(Doppler shift into resonance).

② Resolved sideband cooling (state preparation)

③ Qubit operations.

Resolved Sideband Cooling

While necessary as a cooling step, will also outline how quantum state of ion will be manipulated.

From Hamiltonian above:

$$H_{int} = \frac{\hbar\Omega}{2}\left(e^{-i\omega t + ik\hat{x}} + e^{i\omega t - ik\hat{x}}\right)\left(\sigma_+ + \sigma_-\right)$$

$\frac{-\partial E_0}{\hbar}$ (Rabi frequency or resonance)

$|e\rangle\langle g| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

$|g\rangle\langle e| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$

Translate into rotating frame w/ Unitary

$$U_{RWA} = e^{-i\omega\sigma_z t/2}$$

Exercise: show this

$$U_{RWA}^{+}\sigma_{+}U_{RWA} = e^{i\omega\sigma_z t/2}|e\rangle\langle g|e^{-i\omega\sigma_z t/2}$$

$$= e^{i\omega t}\sigma_{+} \quad \text{(use identity for Pauli operators and } \sigma_z = |g\rangle\langle g| - |e\rangle\langle e|\text{)}$$

$$e^{ix(\hat{n}\cdot\vec{\sigma})} = I\cos x + i(\hat{n}\cdot\vec{\sigma})\sin x$$

Similarly:

$$U_{RWA}^{+}\sigma_{-}U_{RWA} = e^{-i\omega t}\sigma_{-}$$

Transforming $U_{RWA}^{+} H_{int} U_{RWA}$:

$$= \frac{\hbar\Omega}{2}\left(e^{ik\hat{x}}\sigma_{+} + e^{-ik\hat{x}}\sigma_{-}\right) + \frac{\hbar\Omega}{2}\left(e^{2i\omega t}e^{-ik\hat{x}}\sigma_{+} + \text{h.c.}\right)$$

(average to zero $\rightarrow 0$)

Transform remaining state back: (Applies $U_{RWA}\sigma_{+}U_{RWA}^{+}$, etc.)

$$= \frac{\hbar\Omega}{2}\left(e^{-i\omega t + ik\hat{x}}\sigma_{+} + e^{i\omega t + ik\hat{x}}\sigma_{-}\right)$$

To continue on, make some definitions:

$$\hat{x} = \sqrt{\frac{\hbar}{2m\nu}}\left(a^{+} + a\right)$$; where a^{+} a usual creation and anihilation operators.

$$\eta = k\sqrt{\frac{\hbar}{2m\nu}}$$

$k \rightarrow \frac{1}{\lambda}$ where λ is wavelength of light.

$\sqrt{\frac{\hbar}{2m\nu}} \rightarrow$ harmonic oscillator length

$\longrightarrow$ defines Lamb-Dicke parameter; ratio of harmonic oscillator length to wavelength

for ion traps (tightly confined, large ν, large mass) $\eta \ll 1$.

Expand to first order:

$$e^{\pm ik\hat{x}} = e^{\pm i\eta(a^\dagger + a)} \approx 1 \pm i\eta(a^\dagger + a)$$

Plug in:

$$H_{int}^{RWA} \approx \frac{\hbar\Omega}{2}\left(e^{-i\omega t}\sigma_+\left(1 + \eta(a^\dagger + a)\right) + h.c.\right)$$

$$= \frac{\hbar\Omega}{2}\left(e^{-i\omega t}\sigma_+ + e^{i\omega t}\sigma_-\right) + \frac{\hbar\Omega}{2}i\eta\left[(a^\dagger + a)e^{-i\omega t}\sigma_+\right]$$

$$- i\eta\frac{\hbar\Omega}{2}\left[(a^\dagger + a)e^{i\omega t}\sigma_-\right]$$

Another unitary transform:

Interaction picture

$$U = e^{-i\nu a^\dagger a t}e^{-i\omega_0 t\sigma_z/2}; \quad U^\dagger H_{int}^{RWA} U = H_{int,I}^{RWA}$$

Apply this transformation to the relevant parts:

$$U_I^\dagger a U_I = e^{i\nu a^\dagger a t} a e^{-i\nu a^\dagger a t} = \hat{a}(t).$$

Take derivative:

$$\dot{\hat{a}}(t) = i\nu a^\dagger a e^{i\nu a^\dagger a t} a e^{-i\nu a^\dagger a} + e^{i\nu a^\dagger a t} a e^{-i\nu a^\dagger a}(-i\nu a^\dagger a)$$

$$= i\nu U_I^\dagger\left[a^\dagger a, a\right]U_I$$

$$[a^\dagger a, a] = a^\dagger a a - a a^\dagger a$$

$$= [a^\dagger, a] a = -a$$

$$\dot{\tilde{a}}(t) = -i\nu U_I^\dagger a U_I = -i\nu \tilde{a}(t) \Longrightarrow \tilde{a}(t) = e^{-i\nu t} a$$

(just a phase factor)

Apply transformation to σ_+, σ_-

$$U^\dagger \sigma_+ U = e^{i\omega_0 t}\sigma_+ \; ; \; U^\dagger \sigma_- U = e^{-i\omega_0 t}\sigma_-$$

Collect all the terms

$$H_{int}^{RWA} \approx \frac{\hbar\eta}{2}\left(e^{-i\omega t}\sigma_+ e^{i\omega_0 t} + e^{i\omega t}\sigma_- e^{-i\omega_0 t}\right)$$

$$+ \frac{\hbar\eta}{2} i\eta\left[\left(a^\dagger e^{i\nu t} + a e^{-i\nu t}\right) e^{-i\omega t}\sigma_+ e^{i\omega_0 t}\right]$$

$$- \frac{\hbar\eta}{2} i\eta\left[\left(a^\dagger e^{i\nu t} + a e^{-i\nu t}\right) e^{i\omega t}\sigma_- e^{-i\omega_0 t}\right]$$

Define: $\delta = \omega_0 - \omega \qquad \omega = \omega_0 - \delta$

$$H_{int}^{RWA} \approx \frac{\hbar\eta}{2}\left[e^{i\delta t}\sigma_+ + e^{-i\delta t}\sigma_-\right]$$

$$+ i\eta\frac{\hbar\eta}{2}\left[a^\dagger e^{i\nu t} + a e^{-i\nu t}\right] e^{i\delta t}\sigma_+$$

$$- i\eta\frac{\hbar\eta}{2}\left[a^\dagger e^{i\nu t} + a e^{-i\nu t}\right] e^{-i\delta t}\sigma_-$$

Consider Three cases:

① $\delta = 0$ (carrier)

X-gate.

$$H_C = \frac{\hbar\Omega}{2}\left[\sigma_+ + \sigma_-\right] = \frac{\hbar\Omega}{2} X$$

② $\delta = \nu$ (1st red sideband)

$$H_{RSB} = i\eta\frac{\hbar\Omega}{2}\left[a\sigma_+ - a^\dagger\sigma_-\right]$$

③ $\delta = -\nu$ (1st blue sideband)

$$H_{BSB} = i\eta\frac{\hbar\Omega}{2}\left[a^\dagger\sigma_+ - a\sigma_-\right]$$

What do these do to the state $|g, n\rangle$ or $|e, n\rangle$?

approximate $a^\dagger|n\rangle \longrightarrow |n+1\rangle$ (ignore normalisation as we are not too concerned here)

1st case: $\delta = 0$ carrier.

$$[\sigma_+ + \sigma_-]|g,n\rangle = \sigma_+|g,n\rangle + \sigma_-|g,n\rangle$$

$$\sigma_+ = \frac{\sigma_x + i\sigma_y}{2} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} ; \quad \sigma_- = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$\sigma_+|g,n\rangle = \sigma_+|g\rangle \otimes |n\rangle = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes |n\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes |n\rangle = |e,n\rangle$$

$$\sigma_-|g,n\rangle = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$$

So for $\delta = 0$ take $|g,n\rangle \rightarrow |e,n\rangle$

2nd case $\delta = \nu$ (RSB)

$$i\eta\hbar\frac{\Omega}{2}\left[a\sigma_+ - a^\dagger\sigma_-\right]|g,n\rangle \quad (a^\dagger\sigma_- \text{ term} \rightarrow 0)$$

$$= i\eta\hbar\frac{\Omega}{2}|e\rangle \otimes a|n\rangle = i\eta\frac{\hbar\Omega}{2}|e,n-1\rangle$$

So for $\delta = \nu$ take $|g,n\rangle \rightarrow |e,n-1\rangle$

3rd case $\delta = -\nu$ (BSB)

$$i\eta\frac{\hbar\Omega}{2}\left[a^\dagger\sigma_+ - a\sigma_-\right]|g,n\rangle \rightarrow |e,n+1\rangle$$

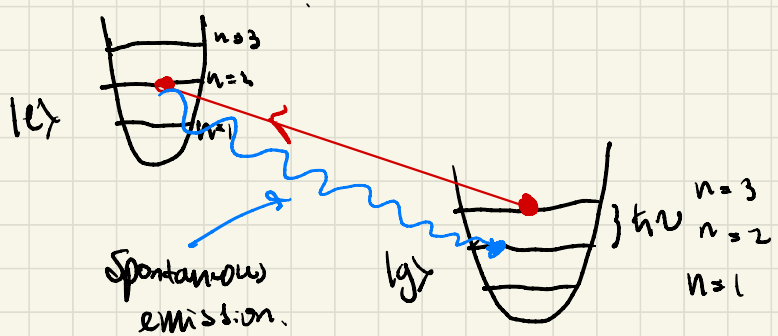
So $\delta = -\nu$ take $|g,n\rangle \rightarrow |e,n+1\rangle$.

Sideband cooling:

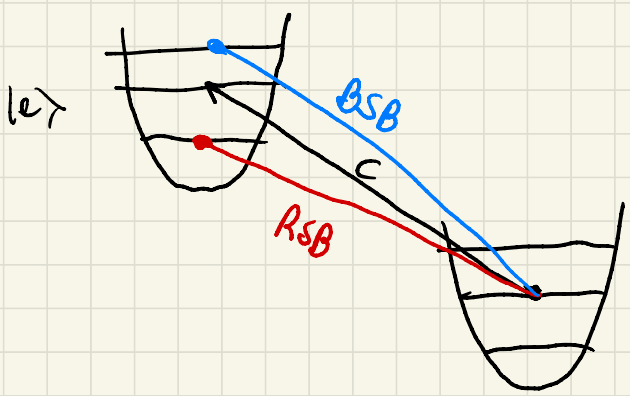
Red side band.

$|g,n\rangle \rightarrow |e,n-1\rangle$; $|e,n-1\rangle \rightarrow |g,(n-1)+1\rangle = |g,n\rangle$

??



General case:



RSB action on $|g,1\rangle$? → Stationary.

$\sigma_+ a |g,1\rangle = 0$ (annihilation operator)

Single qubit rotations (accomplished w/ carrier excitation)

- A more general form of the carrier Hamiltonian (include driving field phase again)

$$H_{carrier} = \frac{\hbar\Omega}{2}\left(e^{i\varphi}\sigma_+ + e^{-i\varphi}\sigma_-\right)$$

$$U(t) = e^{-iH_{carrier}t/\hbar} = ?$$

- Use identity $e^{-i\frac{\Omega t}{2}\sigma_x} = \cos\frac{\Omega t}{2}\mathbb{I} - i\sin\frac{\Omega t}{2}\sigma_x$

Unitary:

$$U(t) = \cos\frac{\Omega t}{2}\mathbb{I} - i\sin\frac{\Omega t}{2}\left(e^{-i\varphi}\sigma_+ + e^{i\varphi}\sigma_-\right)$$

$$U(t)|g\rangle = \cos\frac{\Omega t}{2}|g\rangle - i\sin\frac{\Omega t}{2}e^{-i\varphi}|e\rangle$$

$$U(t)|e\rangle = \cos\frac{\Omega t}{2}|e\rangle - i\sin\frac{\Omega t}{2}e^{i\varphi}|g\rangle$$

Matrix Form:

$$\begin{pmatrix} \cos\frac{\Omega t}{2} & -ie^{-i\varphi}\sin(\Omega t/2) \\ -ie^{i\varphi}\sin(\Omega t/2) & \cos\frac{\Omega t}{2} \end{pmatrix}$$

Same as before → we can do arbitrary rotations on Bloch sphere.

→ compose single qubit gates from these rotations.

"π-pulse", "$\pi/2$ pulse" i.e. $\pi = \Omega t$ (pulse area)

TABLE I. Selected state-of-the-art gate demonstrations.

Gate Type	Gate Method	Fidelity	Gate Time (μs)	Ion Species	Ref.
Single-Qubit					
	Optical	0.99995	5	$^{40}Ca^{+}$	[28]
	Raman	0.99993	7.5	$^{43}Ca^{+}$	[27]
	Raman	0.99996	2	$^{9}Be^{+}$	[37]
	Raman	0.99	0.00005	$^{171}Yb^{+}$	[163]
	Raman	0.999	8	$^{88}Sr^{+}$	[113]
	Microwave	0.999999	12	$^{43}Ca^{+}$	[22]
	Microwave		0.0186	$^{25}Mg^{+}$	[164]
Two-Qubit (1 species)					
	Optical	0.996	–	$^{40}Ca^{+}$	[38]
	Optical	0.993	50	$^{40}Ca^{+}$	[8]
	Raman	0.9991(6)	30	$^{9}Be^{+}$	[37]
	Raman	0.999	100	$^{43}Ca^{+}$	[27]
	Raman	0.998	1.6	$^{43}Ca^{+}$	[40]
	Raman	0.60	0.5	$^{43}Ca^{+}$	[40]
	Microwave (AC B-field gradient)	0.997	3250	$^{43}Ca^{+}$	[165]
	Microwave (DC B-field gradient)	0.985	2700	$^{171}Yb^{+}$	[166]
Two-Qubit (2 species)					
	Raman/Raman	0.998(6)	27.4	$^{40}Ca^{+}/^{43}Ca^{+}$	[167]
	Raman/Raman	0.979(1)	35	$^{9}Be^{+}/^{25}Mg^{+}$	[168]